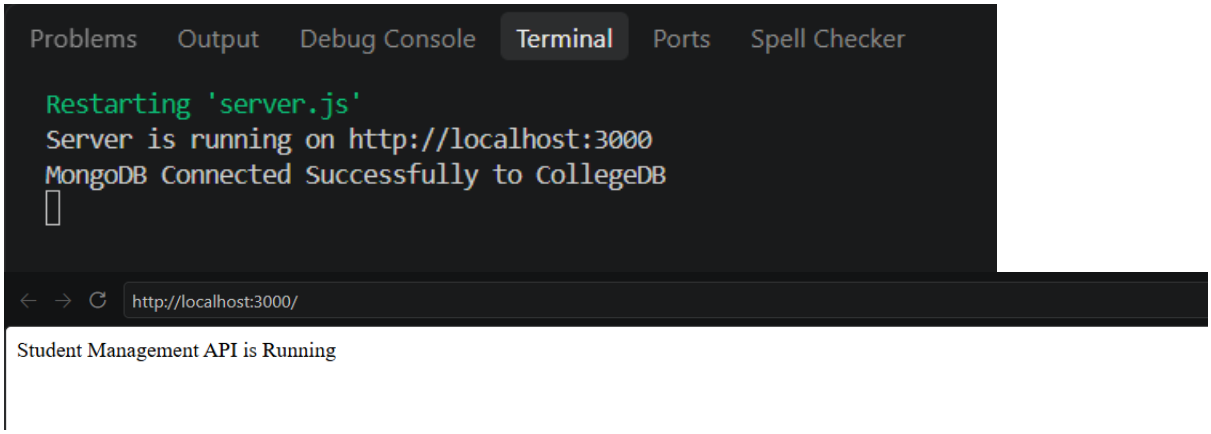
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN STACK
Practical – VII

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Class: TYIT	Batch:1
Date of Assignment: 23-09-26	Date/Time of Submission: 23-09-26

8.API Creation and Integration
a. Create a simple REST API using Express
1) Create a folder & open that in vs code
2) Run -> npm init -y to create package.json file.
3) Add server.js file
4) Run command -> npm install mongoose express cors
5) Add following code into server.js file
Code :

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
.then(() => {
console.log("MongoDB Connected Successfully");
})
.catch((error) => {
console.log("MongoDB Connection Failed");
console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
res.send("Student Management API is Running");
});
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
});
```

Output :



b. Implement a GET API to retrieve data.
Add following code practical (a) code before app.listen()
Code :

```
app.get("/students", async (req, res) => {
try {
const students = await mongoose.connection.db
.collection("students")
.find()
.toArray();
res.json(students);
}
catch (error) {
res.status(500).json({
message: "Error fetching students"
});
}
});
```

Output :



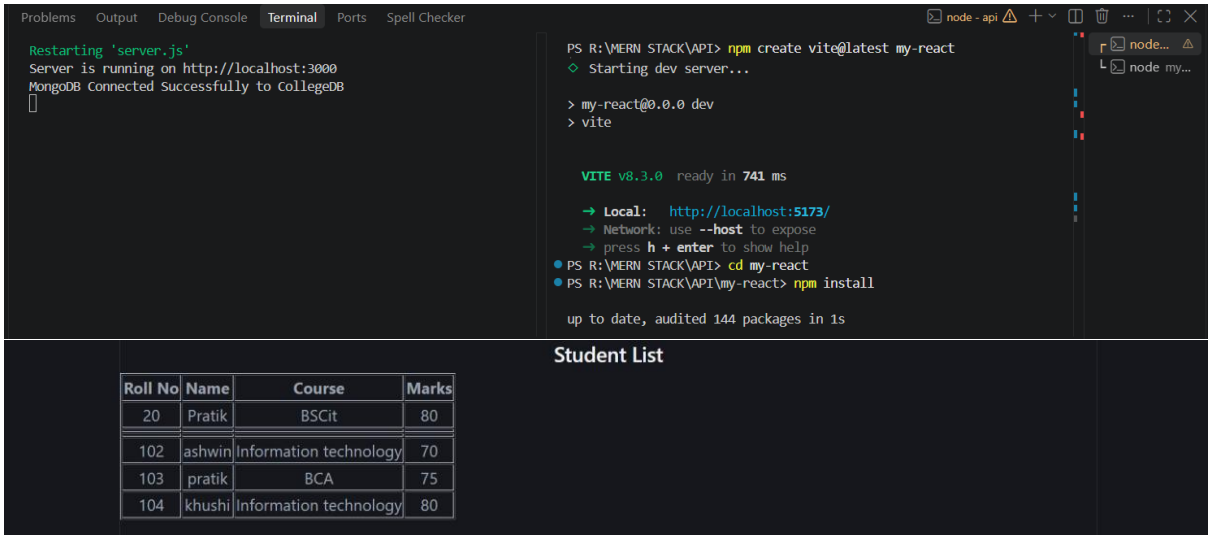
c. Fetch data from the API in a React application.
1)Create react project using following command
npm create vite@latest my-react
2)Choose react, javascript, ESLint
3)In terminal move to my-react & install axios

npm install axios
4)Add following code to App.jsx
Code :

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
const [students, setStudents] = useState([]);
useEffect(() => { axios.get("http://localhost:3000/students")
.then((response) => {setStudents(response.data);})

.catch((error) => {console.log(error);}); }, []);
return (
<div>
<h1>Student List</h1>
<table border="1">
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Course</th>
<th>Marks</th>
</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td>
<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>
</div>
);
}
export default App;
```

5)Split terminal & start server in that terminal. To start execute server.js using command node server.js.
6)Then in 1st terminal execute react application using
npm run dev



d. Send data from React to the server using a POST request.

1)Replace App.jsx code

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
const [students, setStudents] = useState([]);
const [rollNo, setRollNo] = useState("");
const [name, setName] = useState("");
const [course, setCourse] = useState("");
const [marks, setMarks] = useState("");
// Get students
const getStudents = () => {axios.get("http://localhost:3000/students")
.then((response) => {setStudents(response.data);})
.catch((error) => {console.log(error);});
};
useEffect(() => {getStudents();}, []);

// Add student
const addStudent = () => {const newStudent = {
rollNo: Number(rollNo),
name: name,
course: course,
marks: Number(marks)
};
axios.post("http://localhost:3000/students", newStudent)
.then((response) => {alert(response.data.message);
setRollNo("");
setName("");
setCourse("");
setMarks("");
getStudents();
```

```
})
.catch((error) => {
console.log(error);
});
};
return (
<div>
<h1>Student Management System</h1>
<h2>Add Student</h2>
<input
type="number"
placeholder="Roll No"
value={rollNo}
onChange={(e) => setRollNo(e.target.value)}
/>
<br /><br />
<input
type="text"
placeholder="Student Name"
value={name}
onChange={(e) => setName(e.target.value)}
/>
<br /><br />

<input
type="text"
placeholder="Course"
value={course}
onChange={(e) => setCourse(e.target.value)}
/>
<br /><br />
<input
type="number"
placeholder="Marks"
value={marks}
onChange={(e) => setMarks(e.target.value)}
/>
<br /><br />
<button onClick={addStudent}>
Add Student
</button>
<hr />
<h2>Student List</h2>
```

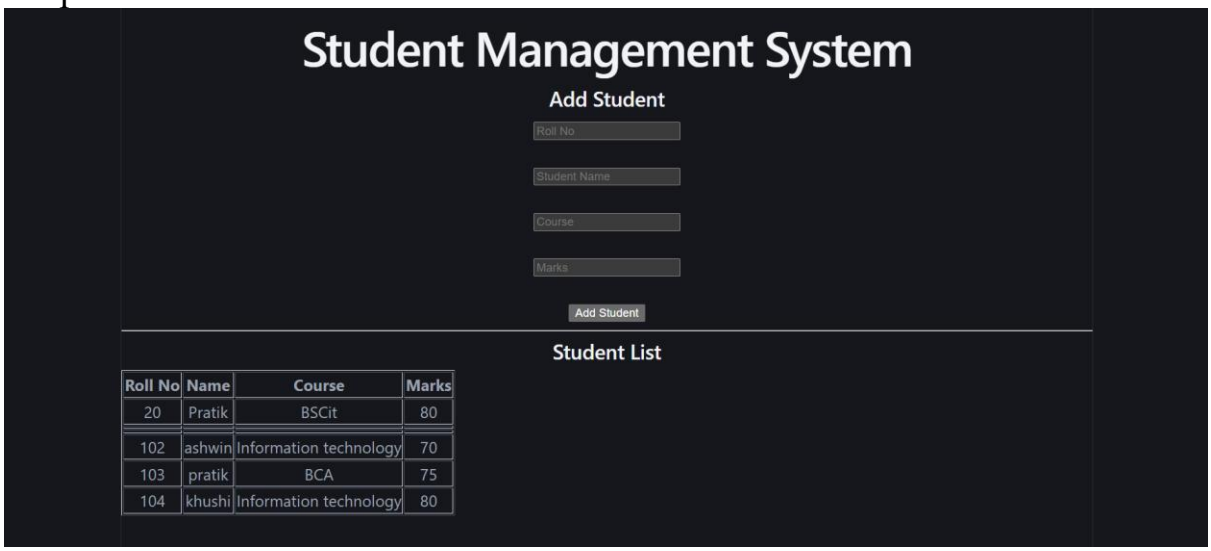
```
<table border="1">
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Course</th>
<th>Marks</th>
</tr>
</thead>
<tbody>
{students.map((student) => (
<tr key={student._id}>
<td>{student.rollNo}</td>
<td>{student.name}</td>
<td>{student.course}</td>
<td>{student.marks}</td>
</tr>
))}
</tbody>
</table>

</div>
);
}
export default App;
```

2)Add post method in server.js file above app.listen()

```
app.post("/students", async (req, res) => {
try {
const student = req.body;
await mongoose.connection.db
.collection("students")
.insertOne(student);
res.json({
message: "Student added successfully",
student: student
});
}
catch (error) {
res.status(500).json({
message: "Error adding student"
});
}
});
```

Output :



e. Store data in MongoDB.

Code :

```
import { useEffect, useState } from "react";
import axios from "axios";

function App() {

   const [students, setStudents] = useState([]);

   const [rollNo, setRollNo] = useState("");
   const [name, setName] = useState("");
   const [course, setCourse] = useState("");
   const [marks, setMarks] = useState("");

   // Get students
   const getStudents = () => {

      axios
         .get("http://localhost:3000/students")
         .then((response) => {
            console.log("Student Data:", response.data);
            setStudents(response.data);
         })
         .catch((error) => {
            console.log("Fetch Error:", error);
         });

   };
```

```
// Fetch data when page loads
useEffect(() => {
   getStudents();
}, []);

// Add student
const addStudent = () => {

   if (!rollNo || !name || !course || !marks) {
      alert("Please fill all fields");
      return;
   }

   const newStudent = {

      rollNo: Number(rollNo),

      name: name,

      course: course,

      marks: Number(marks)

   };

   axios
      .post("http://localhost:3000/students", newStudent)

      .then((response) => {

         alert(response.data.message);

         // Clear form
         setRollNo("");
         setName("");
         setCourse("");
         setMarks("");

         // Refresh student list
         getStudents();

      })
```

```
        .catch((error) => {
            console.log("POST Error:", error);
            alert("Error adding student");
        });
    };
    return (
        <div>
            <h1>Student Management System</h1>
            <h2>Add Student</h2>
            <input
                type="number"
                placeholder="Roll No"
                value={rollNo}
                onChange={(e) => setRollNo(e.target.value)}
            />
            <br />
            <br />
            <input
                type="text"
                placeholder="Student Name"
                value={name}
                onChange={(e) => setName(e.target.value)}
            />
            <br />
            <br />
            <input
                type="text"
                placeholder="Course"
                value={course}
                onChange={(e) => setCourse(e.target.value)}
```

```
/>

<br />
<br />

<input
   type="number"
   placeholder="Marks"
   value={marks}
   onChange={(e) => setMarks(e.target.value)}
/>

<br />
<br />

<button onClick={addStudent}>
   Add Student
</button>

<hr />

<h2>Student List</h2>

<table border="1" cellPadding="10">

   <thead>

      <tr>

         <th>Roll No</th>

         <th>Name</th>

         <th>Course</th>

         <th>Marks</th>

      </tr>

   </thead>

   <tbody>
```

```
                {students.map((student) => (
                  <tr key={student._id}>
                    <td>{student.rollNo}</td>
                    <td>{student.name}</td>
                    <td>{student.course}</td>
                    <td>{student.marks}</td>
                  </tr>
                ))}
              </tbody>
            </table>
          </div>
    );
}

export default App;
```

server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
const PORT = 3000;

// Middleware
app.use(cors());
app.use(express.json());

// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
```

```
    })
    .catch((error) => {
        console.log("MongoDB Connection Failed");
        console.error(error);
    });

// Home API
app.get("/", (req, res) => {
    res.send("Student Management API is Running");
});

// GET API - Fetch students
app.get("/students", async (req, res) => {
    try {
        const students = await mongoose.connection.db
            .collection("Students")
            .find()
            .toArray();

        res.json(students);

    } catch (error) {
        console.error(error);

        res.status(500).json({
            message: "Error fetching students",
            error: error.message
        });
    }
});

// POST API - Store students
app.post("/students", async (req, res) => {
    try {
        const student = req.body;

        await mongoose.connection.db
            .collection("Students")
            .insertOne(student);

        res.json({
            message: "Student added successfully",
            student: student
```

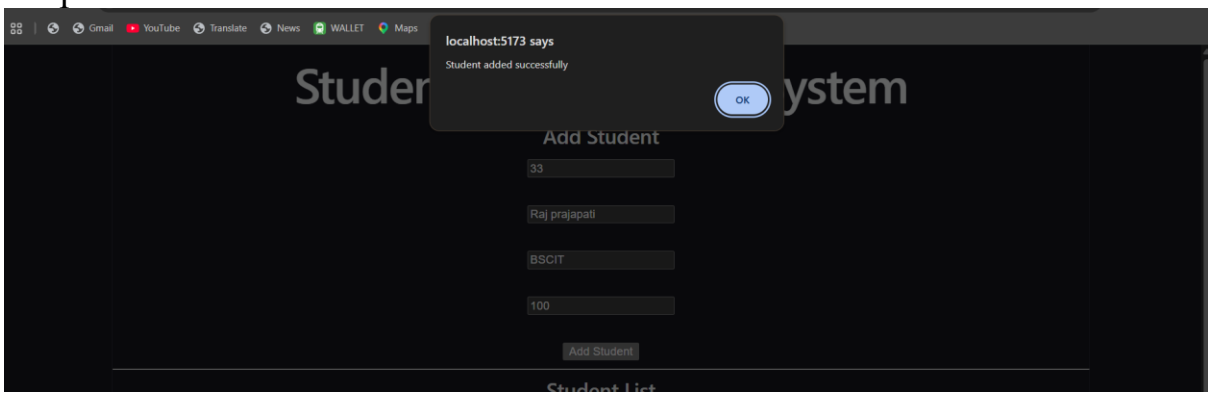
```
        });

    } catch (error) {
        console.error(error);

        res.status(500).json({
            message: "Error adding student",
            error: error.message
        });
    }
});

// Start Server
app.listen(PORT, () => {
    console.log(`Server is running on http://localhost:${PORT}`);
});
```

Output :



f. Display stored data in React page.

